

PCB Design Checklist

Schematic Verification

- ☒ Symbols verified
 - ☒ Component values verified
 - ☒ Net labels verified
 - ☒ Power flags added
 - ☒ ERC completed successfully
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PCB Layout

- ☒ Board outline completed
 - ☒ Components placed logically
 - ☒ Battery polarity verified
 - ☒ Connector orientation verified
 - ☒ Reference designators readable
 - ☒ Silkscreen checked
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Routing

- ☒ Power traces widened
 - ☒ Signal traces separated
 - ☒ No acute routing angles
 - ☒ Analog traces minimized
 - ☒ Ground plane added
 - ☒ Thermal relief enabled
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Verification

- ☒ Design Rule Check passed
 - ☒ No unconnected nets
 - ☒ No clearance violations
 - ☒ No overlapping footprints
 - ☒ Copper pour verified
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Manufacturing Files

- ☒ Gerber files generated
 - ☒ Drill files generated
 - ☒ Gerber Viewer verification completed
 - ☒ Bill of Materials generated
 - ☒ Fabrication package prepared
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Documentation

- ☒ README completed
 - ☒ Project report completed
 - ☒ Design decisions documented
 - ☒ Manufacturing notes documented
 - ☒ Component datasheets included
 - ☒ Images exported
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Final Status

Project Name: Li-Ion Battery Protection & Voltage Monitoring PCB

Design Software: KiCad 9

PCB Size: 90 mm × 40 mm

Layers: 2

Design Rule Check: Passed (0 Errors)

Manufacturing Status: Ready for fabrication